

# Calibration Report Final

## Overview

This report analyzes longitudinal calibration data to understand the typical, variable, stationary, and nonstationary behavior of calibration series.

The overall workflow is:

1. Setup: Load and clean the calibration data
2. Stationarity Testing
3. White Noise and Stationary Classification using Ljung-Box and Augmented Dickey-Fuller Tests
4. Applying Moving Average Smoothing for “Ground Truth”
5. Interpretation: Determining a Calibration Threshold
6. Next Steps

# 1. Setup

## 1.1 Data Loading

Loads required R packages, imports the geographic landmark metadata, and reads calibration CSV files containing daily calibration rates from all anchors between January 6th, 2026 and April 14th, 2026.

## 1.2 Cleaning Function

The calibration files contain anchor ID and calibration rate information in slope “m” and intercept “b” values. The cleaning function reallocates the automatically read-in column names as the first row, separates the second column into slope and intercept columns, and combines the two numeric components into a single calibration value by using the  $y = mx + b$  equation with an arbitrary  $x = 100$ . The resulting col4 variable is  $y$ , the calibration value used throughout the remainder of the analysis.

### 1.2.1 Keep IDs Common Across Calibration Files

Not every calibration ID necessarily appears in every file. To choose a sample of anchors that are sufficiently represented across daily measurements, an anchor is retained when its ID appears in at least 80% of the available files. This reduces the likelihood that missing observations dominate time series analysis and across-anchor behavior. 900 of 961 of anchors are retained.

### 1.2.2 Build the Longitudinal Calibration Dataset

The calibration CSVs are reduced to anchor ID and daily calibration values, with rows corresponding to calibration IDs and columns corresponding to days. There are 98 days of sequential calibration data, but due to days with large volumes of missing data, 18 days were omitted. Namely, a dozen days of calibration data were omitted from the beginning of January, and the remaining days were omitted between January 30th and April 14th, the final day of calibration data collection. Years worth of daily data would be ideal for this study, but we introduce our first calibration rate observations with a three month period. Among the 900 anchor IDs that appeared in at least 80% of our data, only IDs without missing values across all 80 remaining days are retained, leaving 444 of 900 anchor IDs for analysis. The wide calibration dataset is converted to long format so that each row represents one ID-day observation, and so that only one dataframe must be read in for plotting visualizations. Geographic landmark information is then joined to each calibration series—allowing later analyses to examine calibration behavior by continent, country, or ASN.

## 2. Stationarity Testing

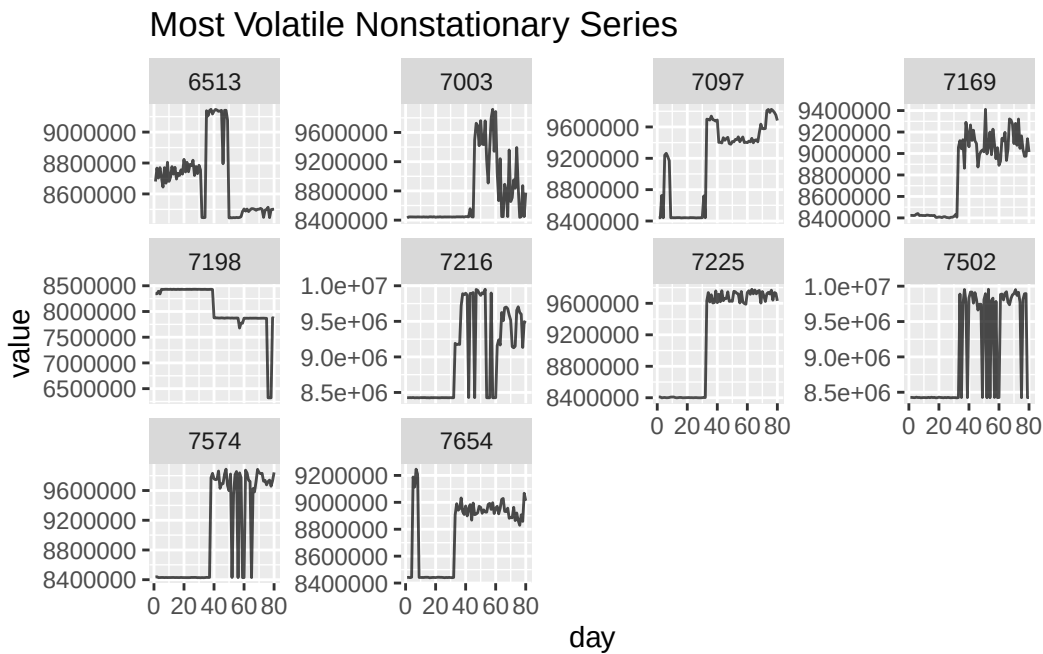
The Augmented Dickey-Fuller (ADF) test is applied independently to every calibration series. A p-value below 0.05 is classified as stationary. This allows the analysis to separate calibration series with relatively stable statistical behavior from series whose underlying behavior changes over time.

```
# A tibble: 1 x 2
  stationary nonstationary
    <int>      <int>
1      179      265
```

### 2.1 Volatility of Nonstationary Series

The next step identifies the most highly volatile nonstationary calibration series. Volatility is measured as the mean absolute change between consecutive observations. Standard deviation is also calculated as a second measure of overall variation.

The ten most volatile nonstationary series are plotted below:



### 3. White Noise and Stationarity Classification

The next stage distinguishes random-looking calibration series from series with detectable temporal structure. The Ljung-Box test is used to identify white-noise behavior, while the ADF test is used to classify stationarity.

The two tests are combined into four initial categories: White noise & stationary, White noise & nonstationary, Structured & stationary, and Structured & nonstationary

```
# A tibble: 4 x 3
  category          n  prop
  <chr>          <int> <dbl>
1 Nonstationary (structured)    225 0.507
2 Stationary (structured)       77 0.173
3 White noise (nonstationary?)   40 0.0901
4 White noise (stationary)     102 0.230
```

#### 3.1 White-Noise Classification

The Ljung-Box test is also isolated so that white-noise series can be analyzed independently of the stationarity classification.

```
# A tibble: 1 x 3
  total white_noise proportion_white_noise
  <int>      <int>          <dbl>
1   444      142          0.320
```

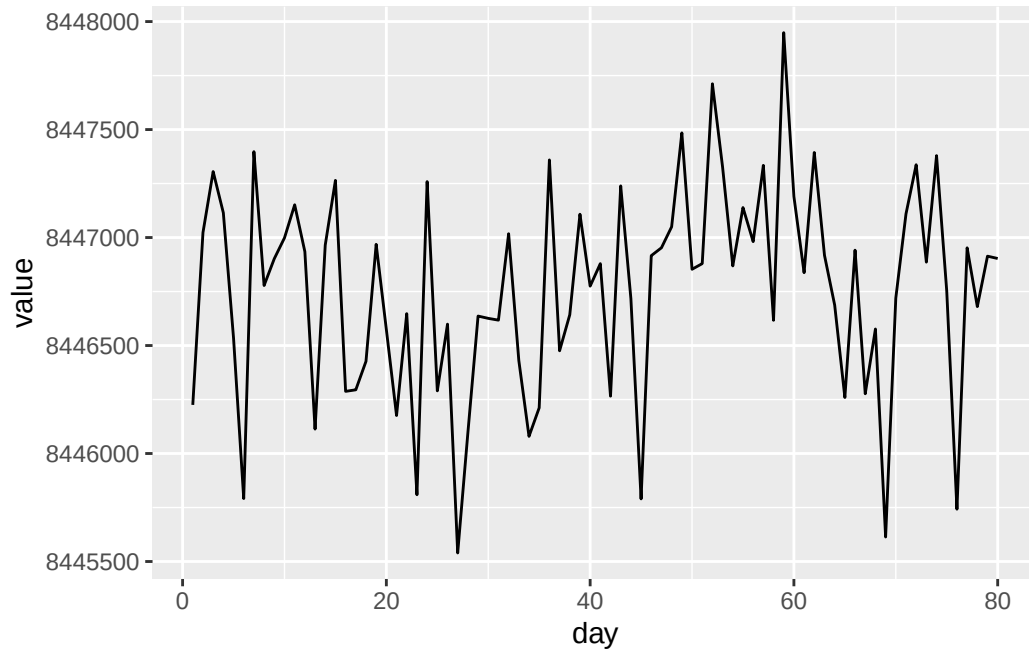
#### 3.2 Stationarity Among Non-White-Noise Series

For series that are not classified as white noise, the ADF test is applied again to explicitly identify which of those series are stationary.

This produces the final three-way classification: White noise (unstructured), Stationary (structured), and Nonstationary (structured).

### 3.3 Investigating White-Noise but Nonstationary Series

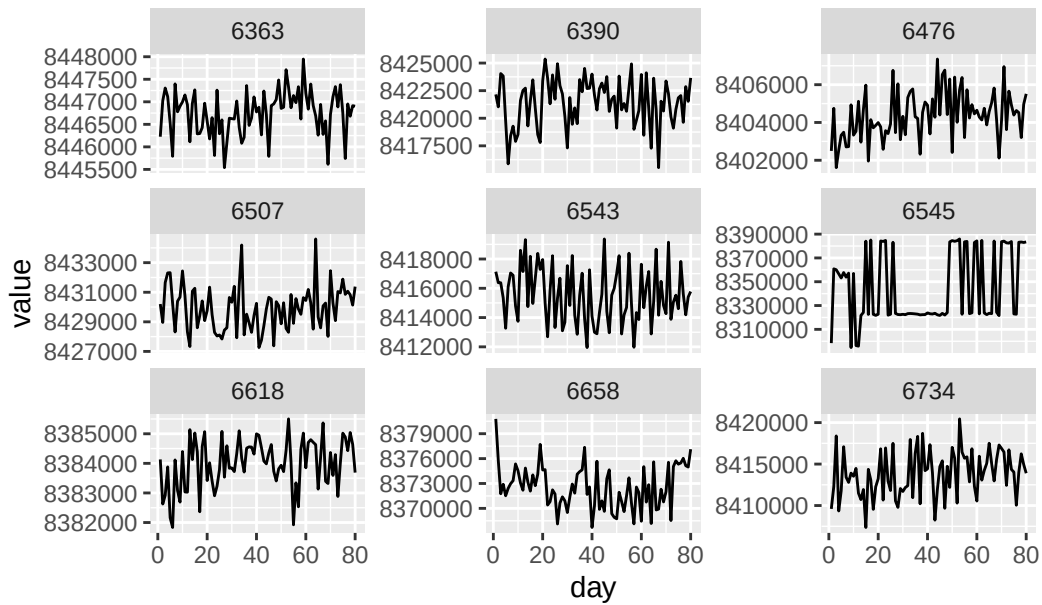
Some series are classified as white noise by the Ljung-Box test but nonstationary by the ADF test. A random ID from the category is rendered below:



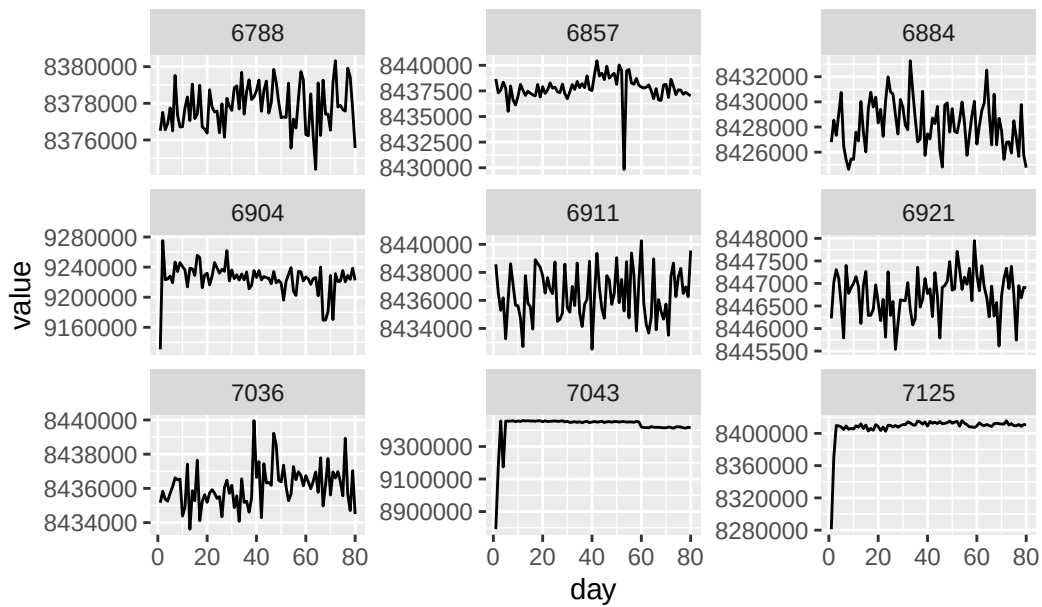
### 3.4 Visualize All White-Noise / Nonstationary Series

All white-noise/nonstationary series are plotted in groups of nine to allow visual inspection of potential drift, steps, variance changes, or other instability.

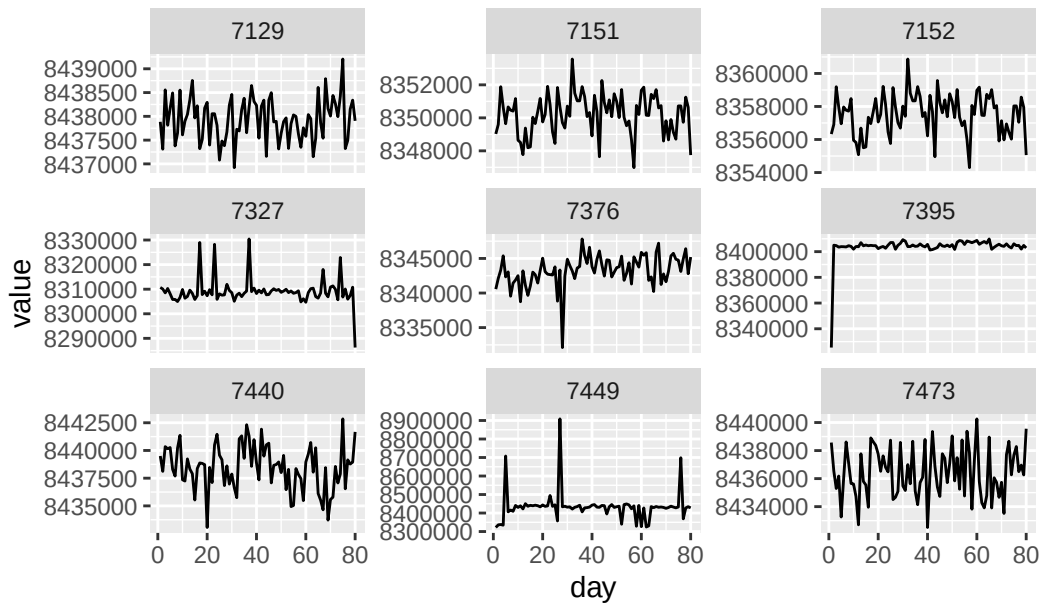
## White Noise / Nonstationary — Page 1



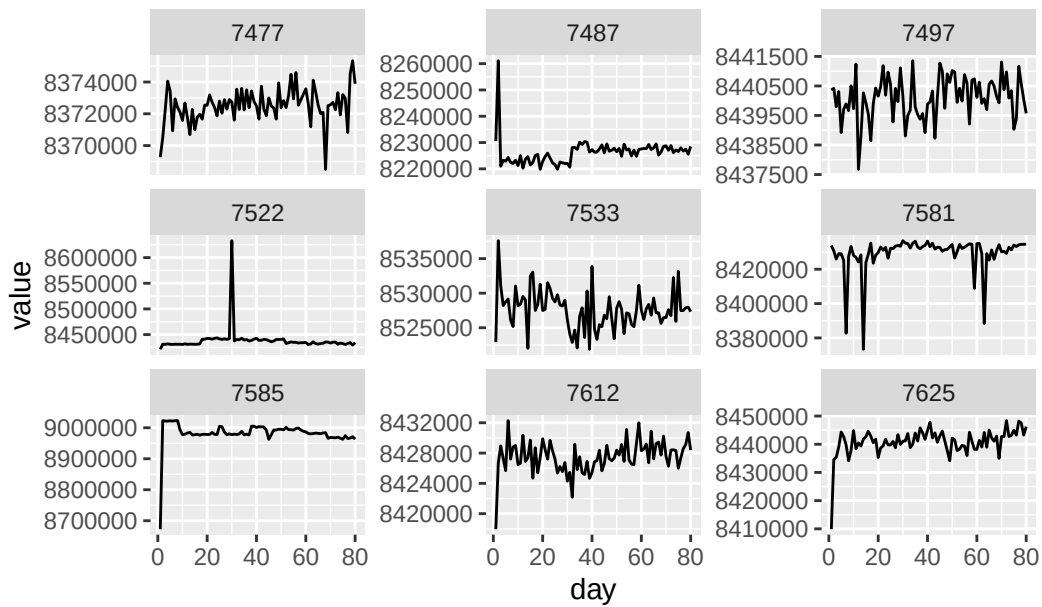
## White Noise / Nonstationary — Page 2



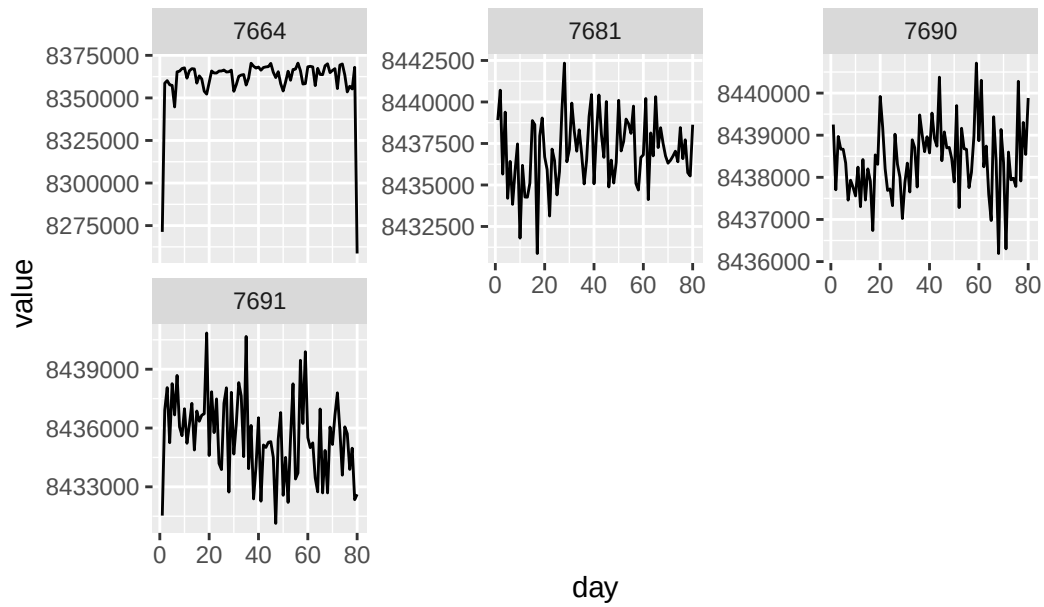
# White Noise / Nonstationary — Page 3



# White Noise / Nonstationary — Page 4



## White Noise / Nonstationary — Page 5



### 3.4.1 Save White-Noise / Nonstationary Visualizations

The same visualizations are saved as PNG files for later inspection.



## 4. Moving-Average

### 4.1 Moving-Average Stationarity Check

The white-noise/nonstationary series are smoothed using a centered MA(5). The purpose is to reduce short-term fluctuations and test whether the underlying smoothed signal behaves more like a stationary process.

### 4.2 Compare Original and Smoothed Stationarity

The original ADF classification is compared with the ADF result after MA(5) smoothing. This determines how many white-noise/nonstationary series become stationary after short-term fluctuations are smoothed out.

```
# A tibble: 40 x 4
  ID      adf_p ma_adf_p ma_stationary
  <chr>  <dbl>    <dbl> <lgl>
1 6363  0.186    0.227 FALSE
2 6390  0.0673   0.01  TRUE
3 6476  0.0553   0.317 FALSE
4 6507  0.0577   0.0113 TRUE
5 6543  0.108    0.341 FALSE
6 6545  0.0854   0.0690 FALSE
7 6618  0.0552   0.0266 TRUE
8 6658  0.655    0.383 FALSE
9 6734  0.0993   0.01  TRUE
10 6788  0.130    0.176 FALSE
# i 30 more rows
```

### 4.3 First Smoothing Stage: Summary of MA(5) Results

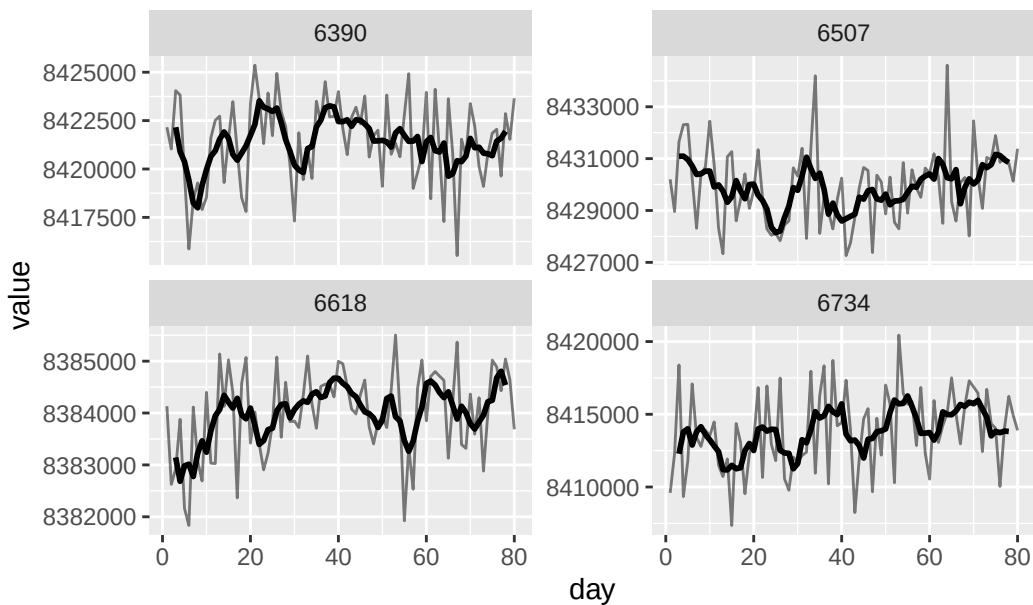
The number and proportion of initially nonstationary white-noise series that become stationary after MA(5) smoothing are calculated.

```
# A tibble: 1 x 3
  total become_stationary proportion
  <int>          <int>          <dbl>
1    40             16           0.4
```

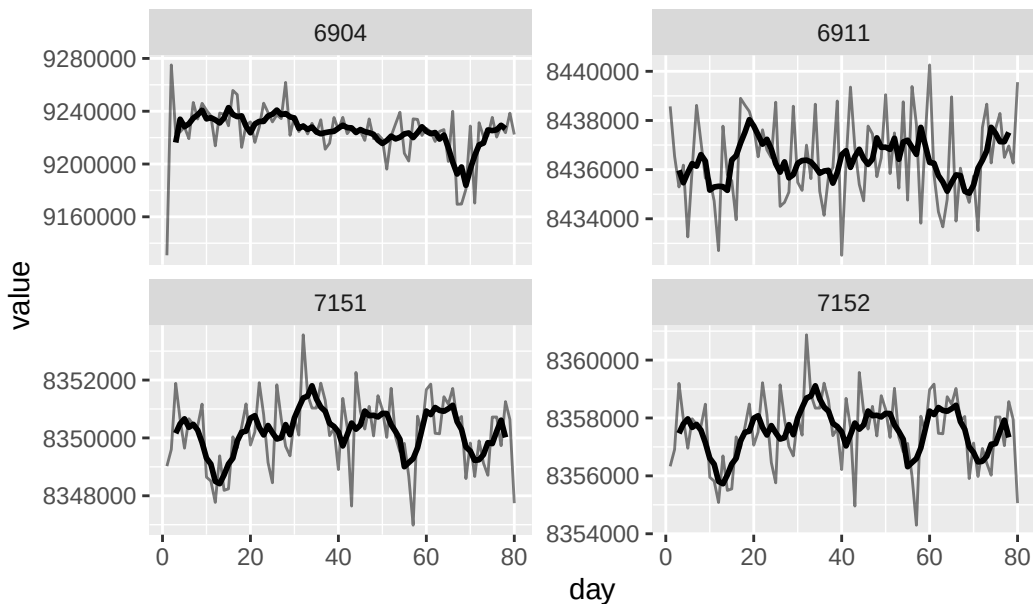
### 4.3.1 Visualize Series Becoming Stationary After MA(5)

Series that become stationary after MA(5) are visualized with both the original and smoothed signals.

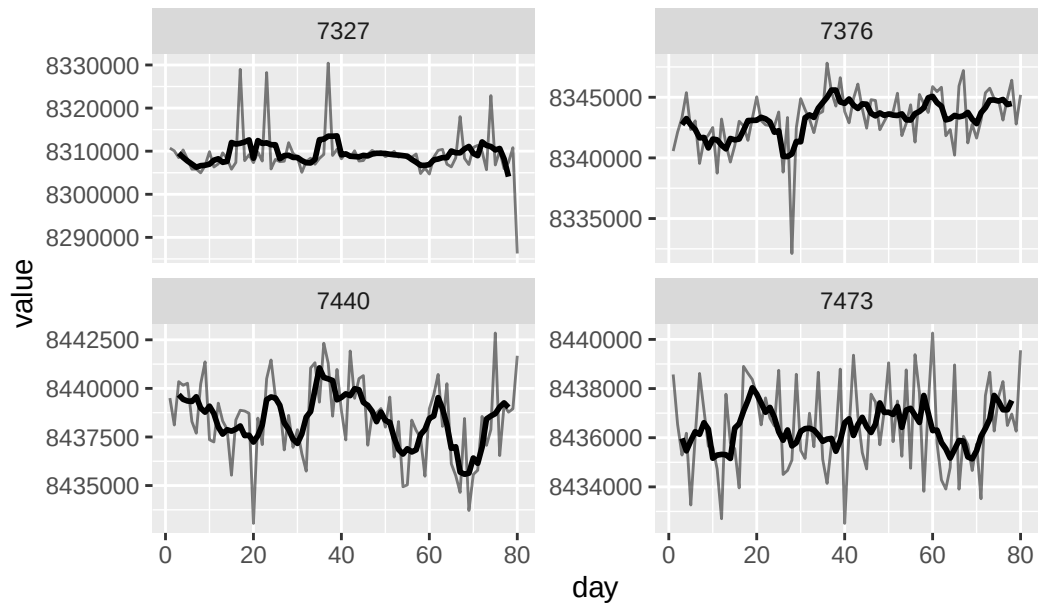
MA-Stationary Series — Page 1



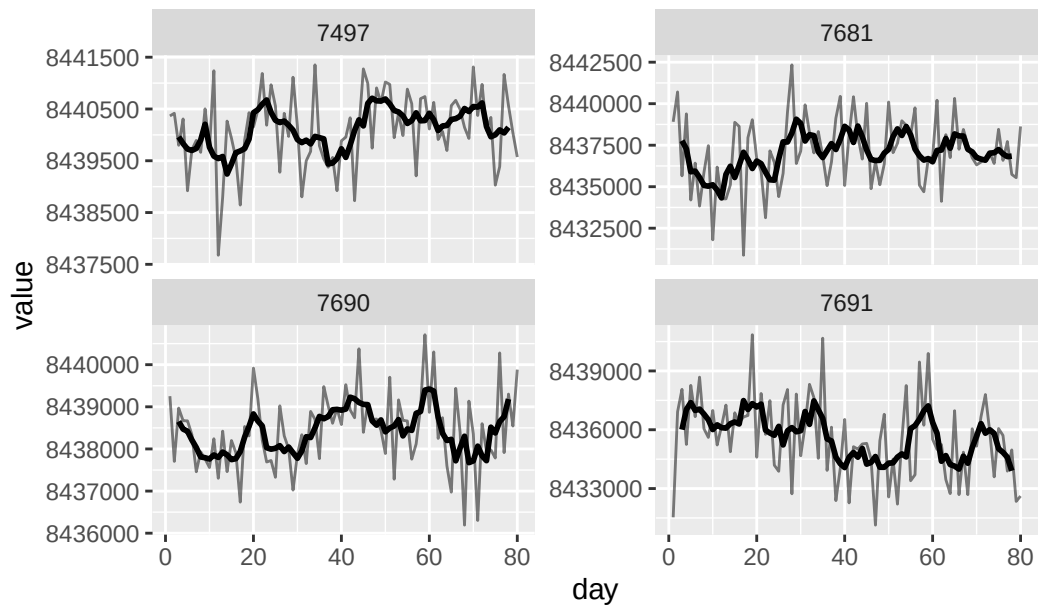
MA-Stationary Series — Page 2



### MA-Stationary Series — Page 3



### MA-Stationary Series — Page 4



#### 4.3.2 Save MA(5) Visualizations

The MA(5) plots are saved as PNG files.

### 4.3.3 Remaining White-Noise / Nonstationary Series

The series that remain nonstationary after the first MA(5) smoothing are isolated for additional smoothing.

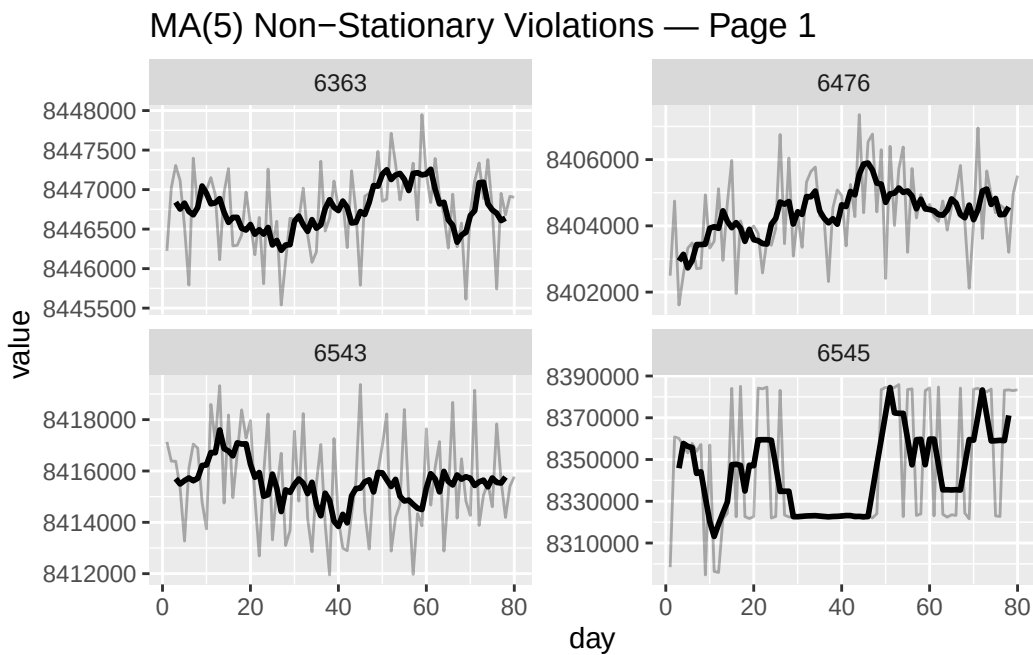
[1] 24

## 4.4 Second Smoothing Stage: Summary of MA(5) → MA(10) Results

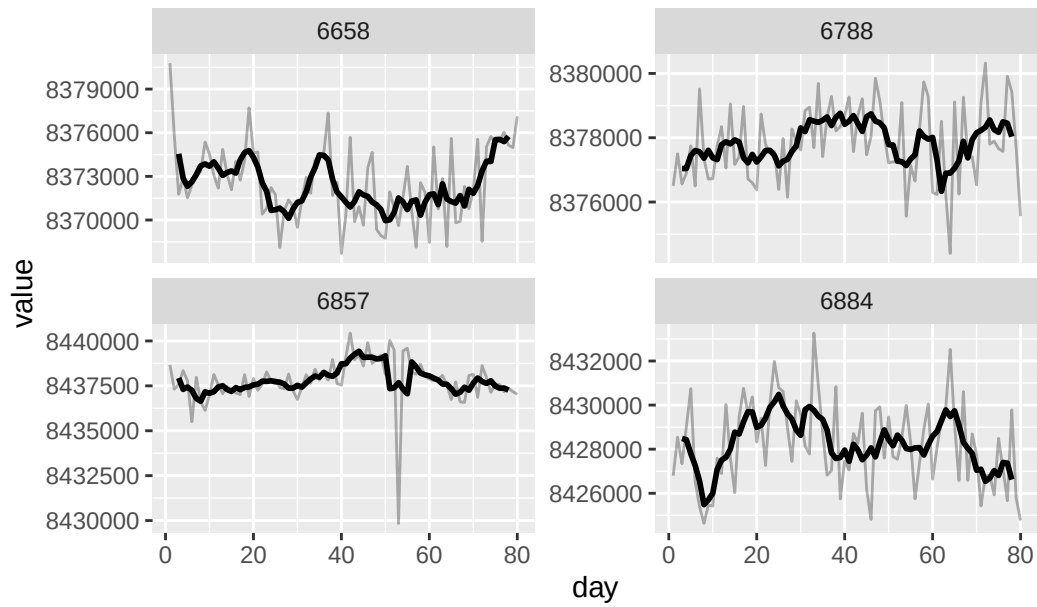
The remaining series are subjected to a second smoothing stage. The first MA(5) reduces short-term variation, and an MA(10) is then applied to the resulting signal. The resulting smoothed series is tested for stationarity again.

### 4.4.1 Visualize Remaining MA(5) Violations

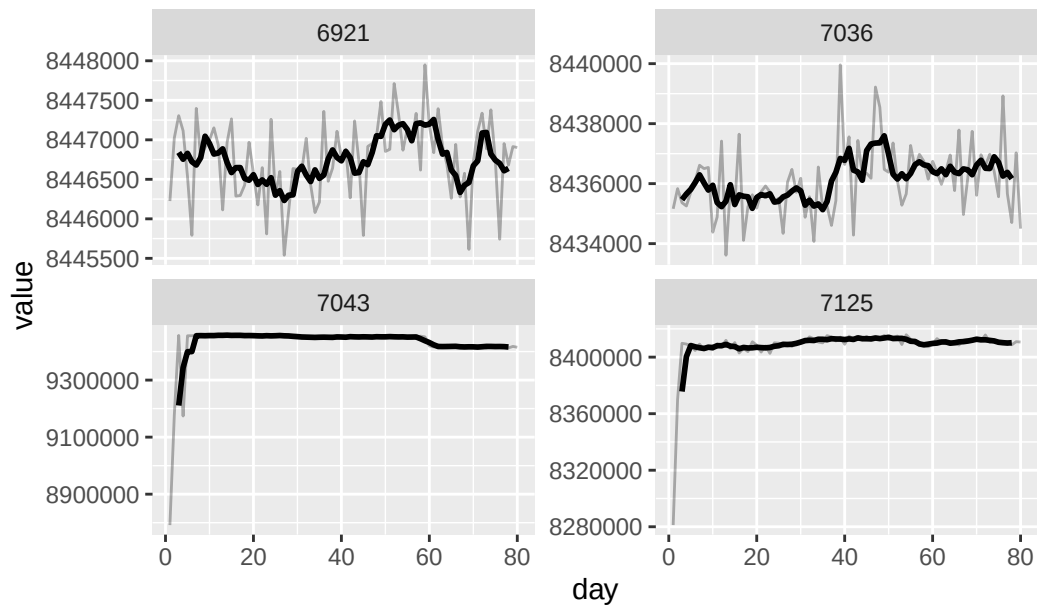
The series that remain nonstationary after the MA(5) stage are plotted with both the original signal and its smoothed version.



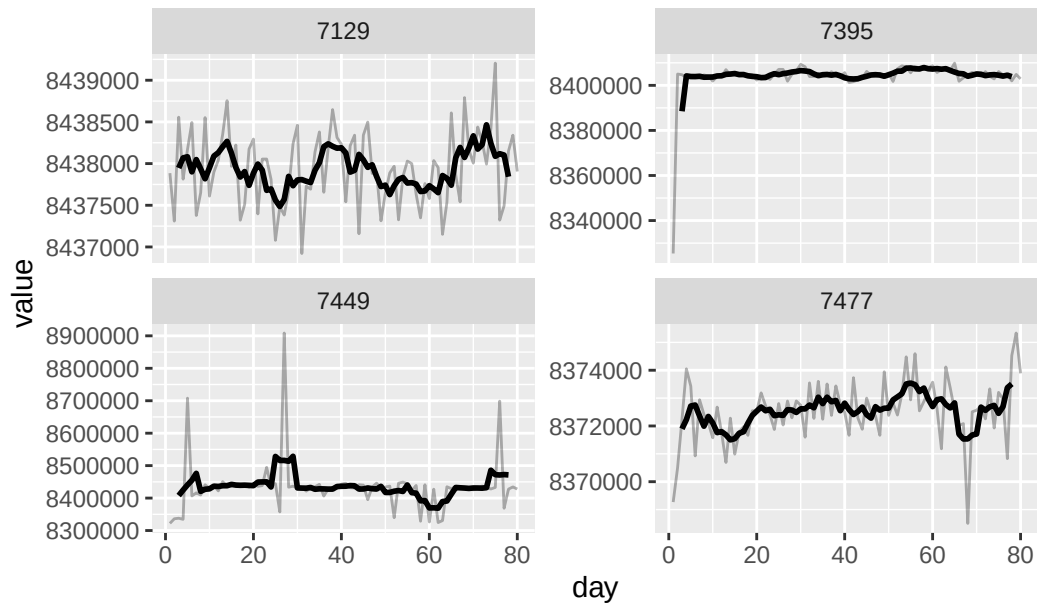
## MA(5) Non-Stationary Violations — Page 2



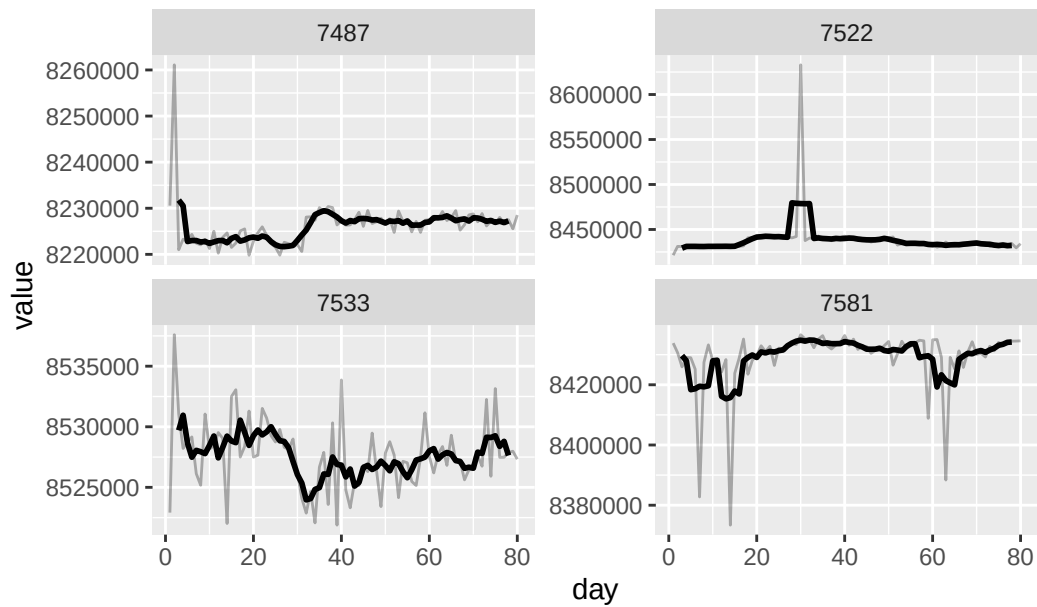
## MA(5) Non-Stationary Violations — Page 3



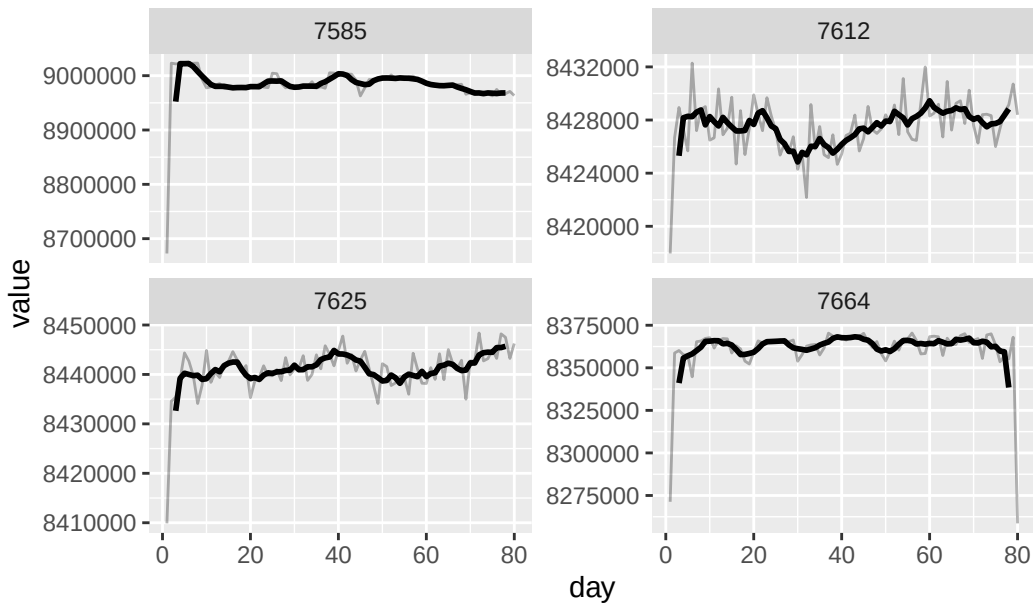
## MA(5) Non-Stationary Violations — Page 4



## MA(5) Non-Stationary Violations — Page 5



## MA(5) Non-Stationary Violations — Page 6



### 4.4.2 Double Moving Average: MA(5) → MA(10)

The remaining nonstationary series are smoothed again using an MA(10) applied to the MA(5) signal. This is intended to further isolate slower-moving calibration behavior from short-term fluctuations.

### 4.4.3 Test Stationarity After Double Smoothing

The MA(10)-on-MA(5) signal is tested using the ADF test.

### 4.4.4 Summary of Double-Smoothing Results

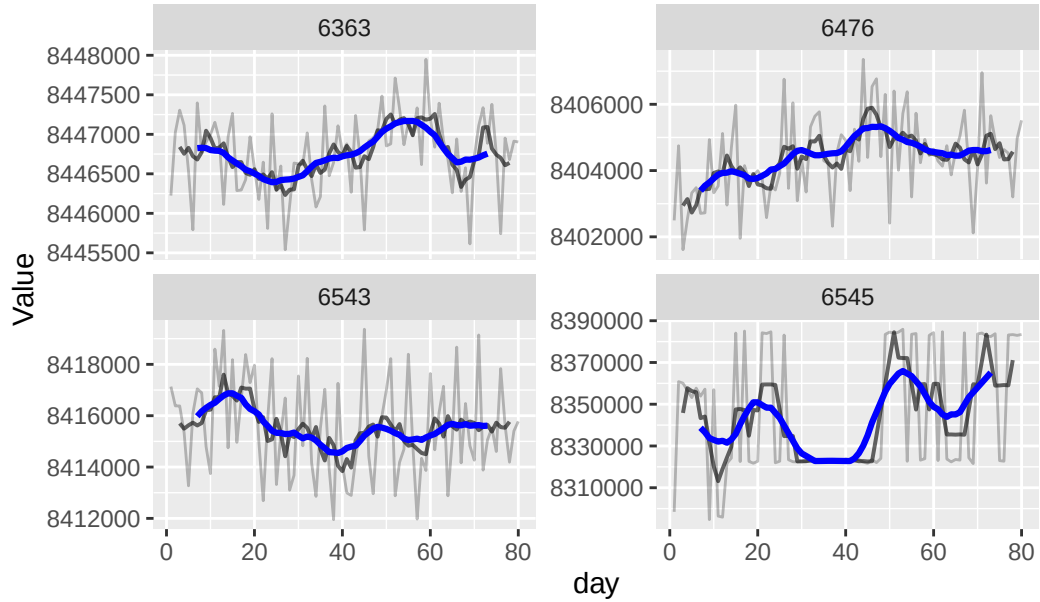
The proportion of the remaining series that become stationary after the second smoothing stage is calculated.

```
# A tibble: 1 x 3
  total stationary_after_double_ma proportion
  <int>          <int>          <dbl>
1     24             7         0.292
```

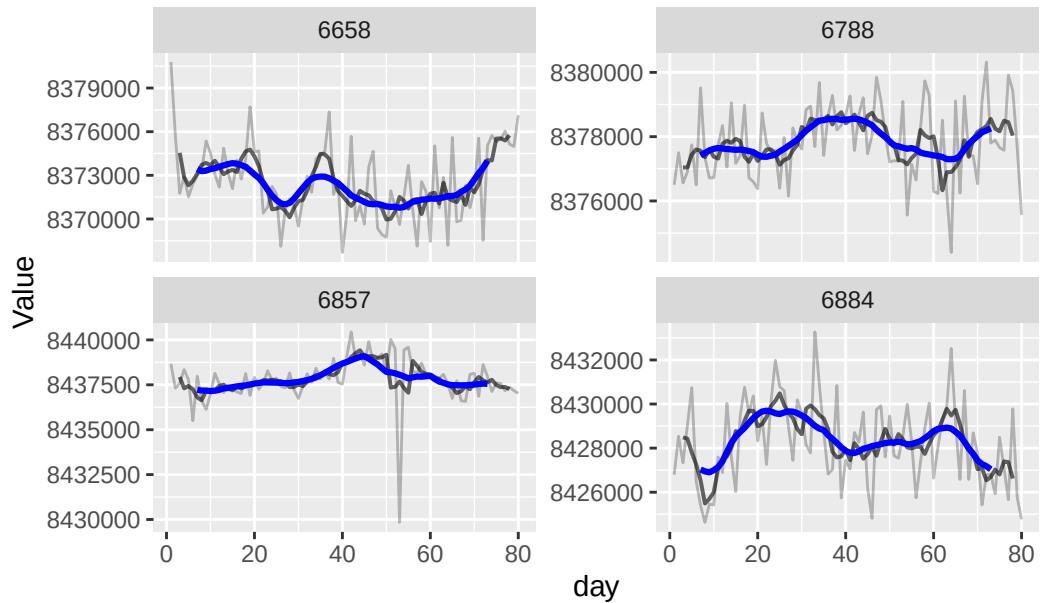
#### 4.4.5 Visualize Double-Smoothing Results

The original signal, MA(5), and MA(10)-on-MA(5) signals are plotted together.

Double Moving Average Analysis (MA5 ... MA10) — Page 1

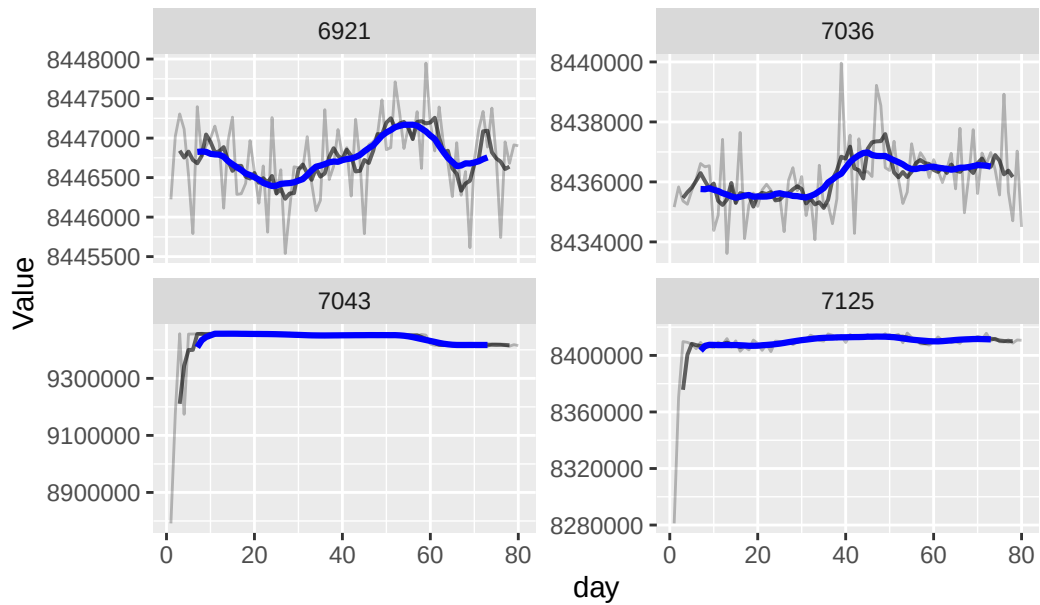


Double Moving Average Analysis (MA5 ... MA10) — Page 2

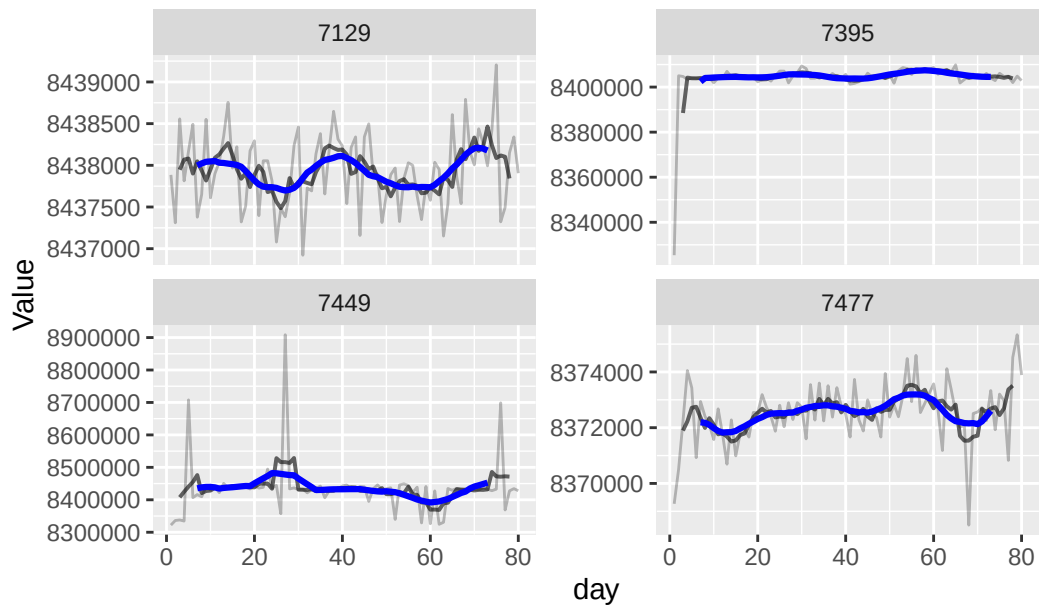




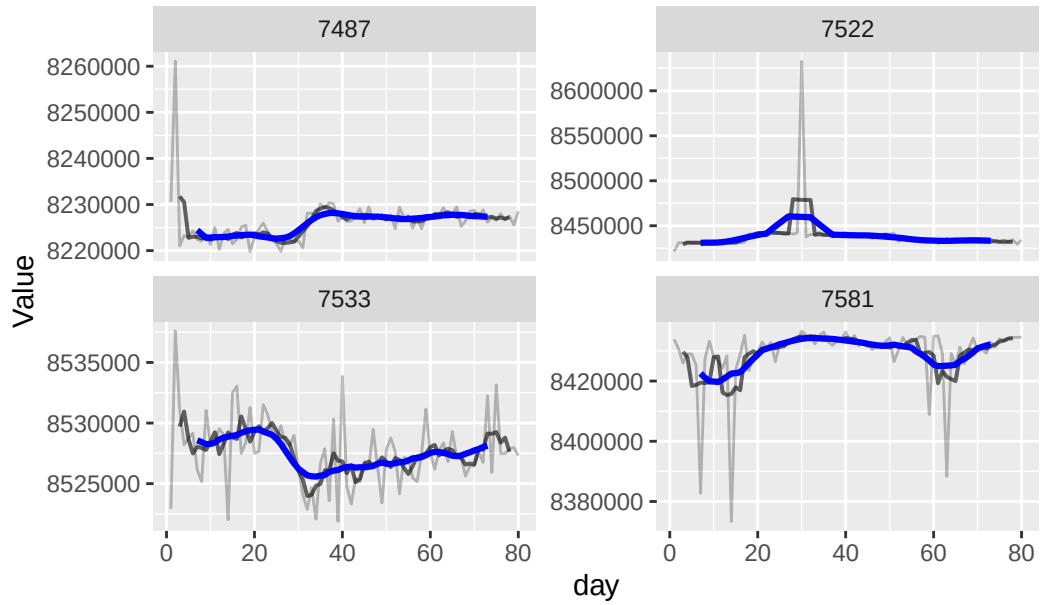
## Double Moving Average Analysis (MA5 ... MA10) — Page 3



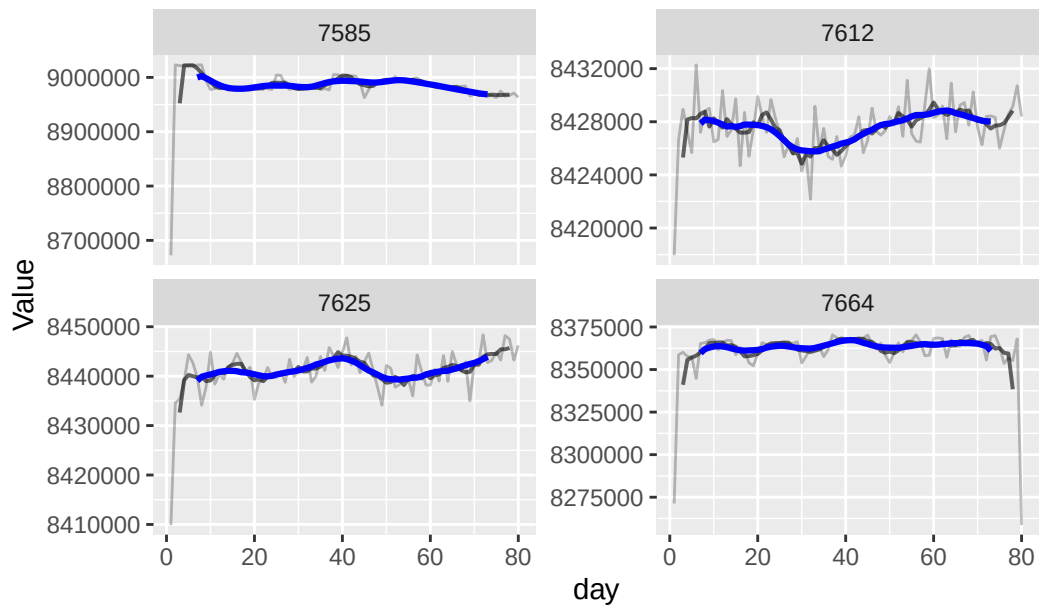
## Double Moving Average Analysis (MA5 ... MA10) — Page 4



## Double Moving Average Analysis (MA5 ... MA10) — Page 5



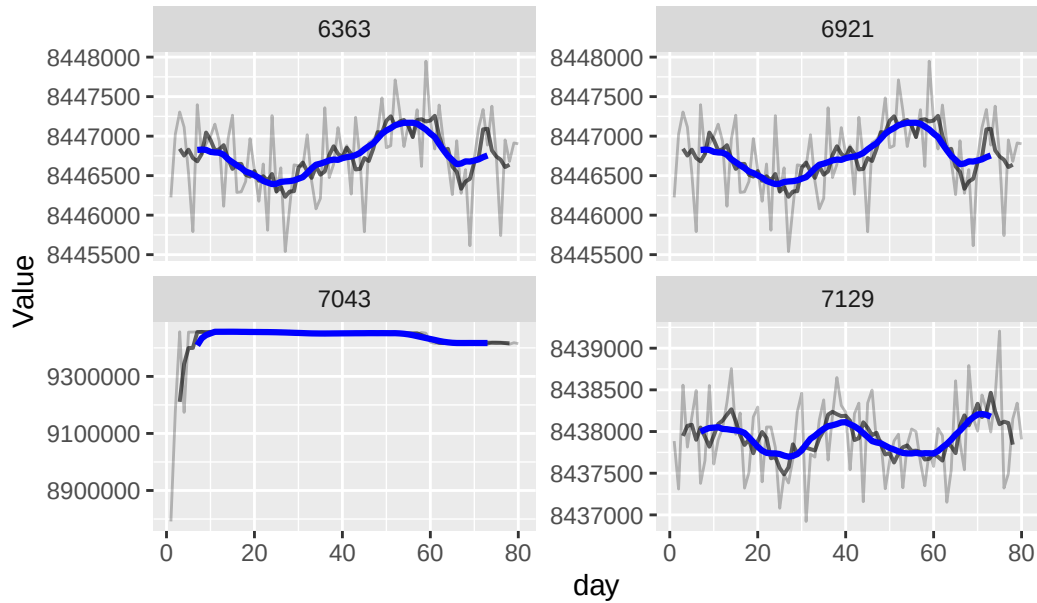
## Double Moving Average Analysis (MA5 ... MA10) — Page 6



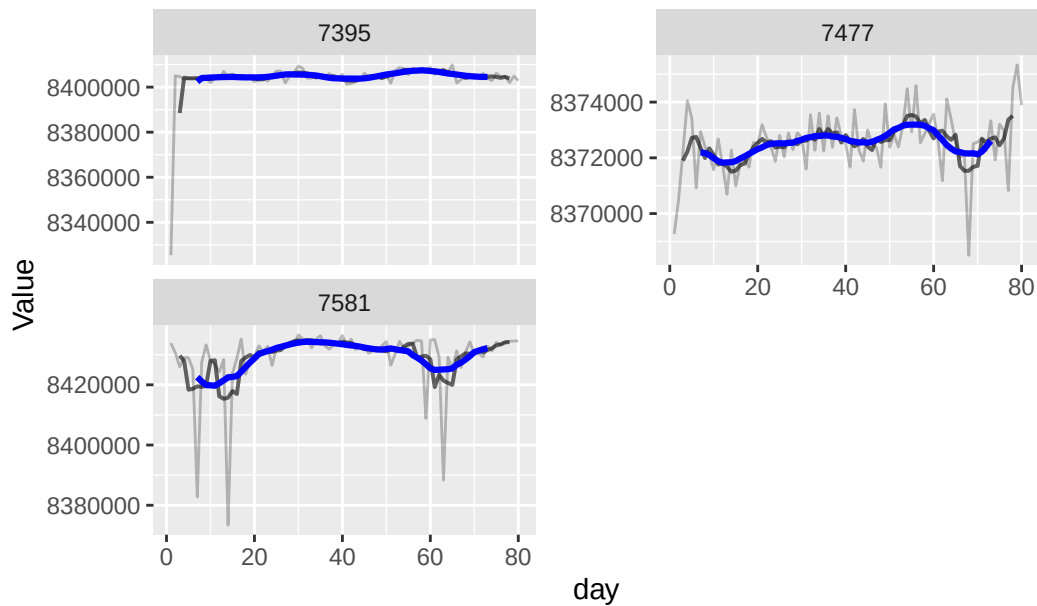
#### 4.4.6 Series Becoming Stationary After Double Smoothing

The IDs that become stationary after the MA(5)  $\rightarrow$  MA(10) process are isolated and visualized separately.

Series Becoming Stationary After MA(5 ... MA10) — Page



Series Becoming Stationary After MA(5 ... MA10) — Page



## 4.5 Third Smoothing Stage: Summary of MA(5) → MA(10) → MA(4) Results

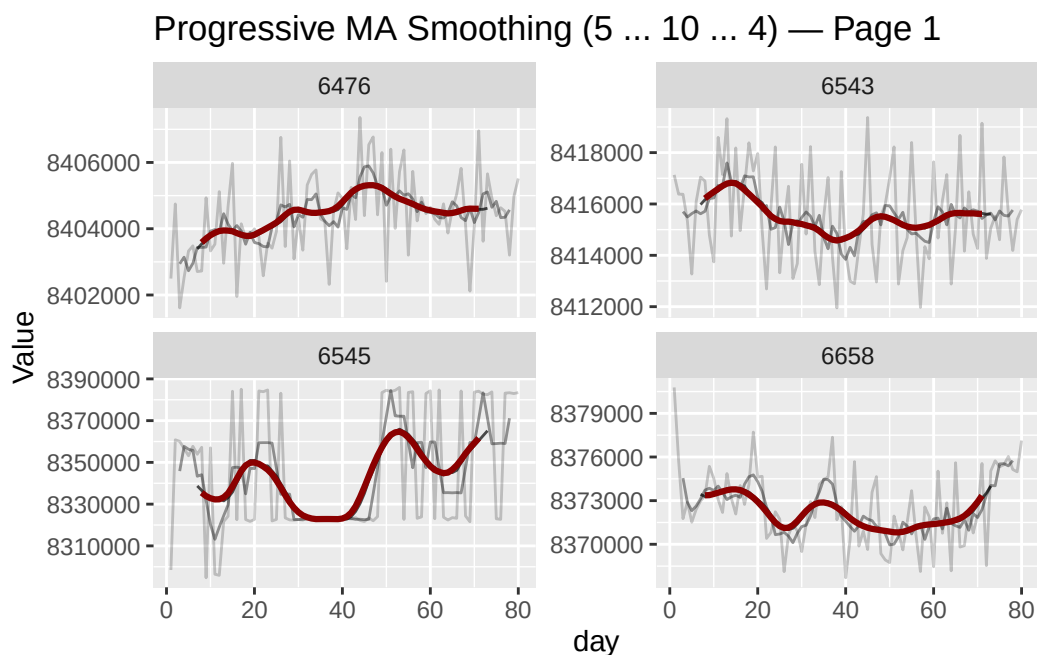
The series that remain nonstationary after the first two smoothing stages are isolated. A final MA(4) is applied to the already smoothed MA(10)-on-MA(5) signal.

### 4.5.1 Test Stationarity After Third Smoothing

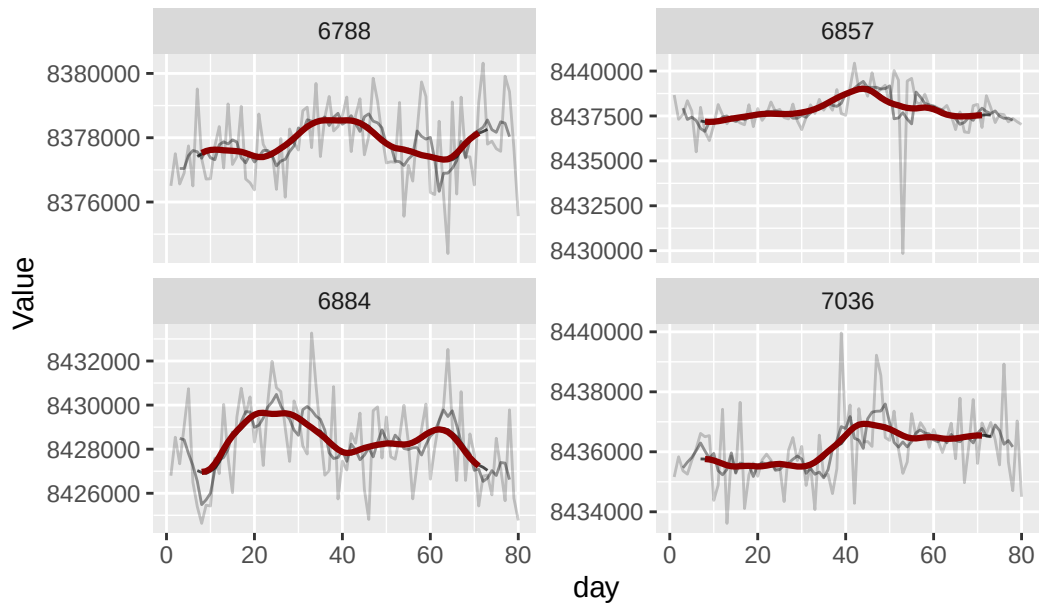
The final smoothed signal is tested for stationarity using the ADF test.

### 4.5.2 Visualize Progressive Smoothing

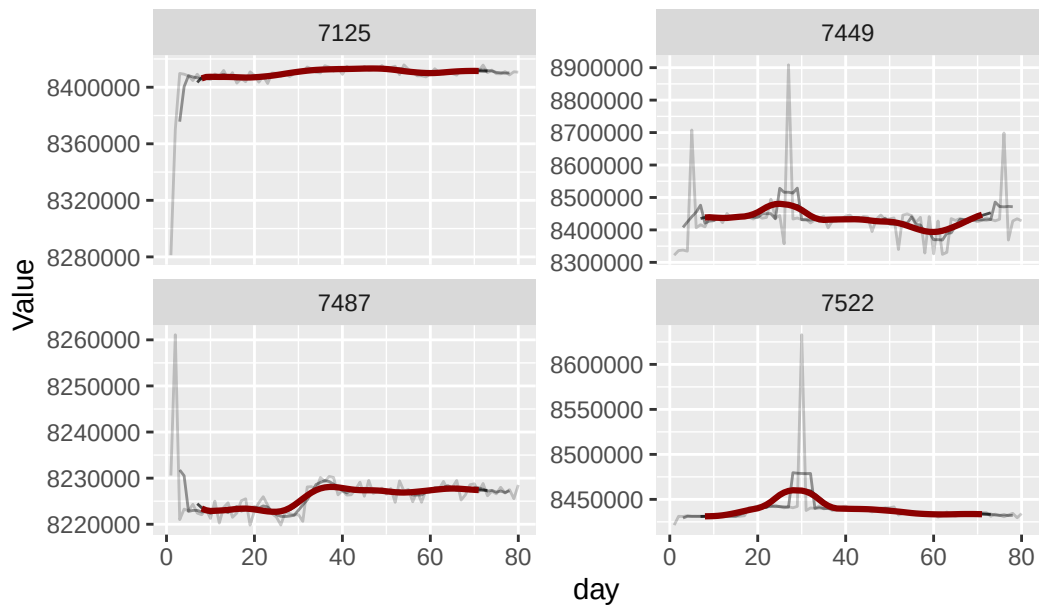
The original signal and all three smoothed versions are displayed together: original, MA(5), MA(10) applied to MA(5), and MA(4) applied to the previous smoothed signal.



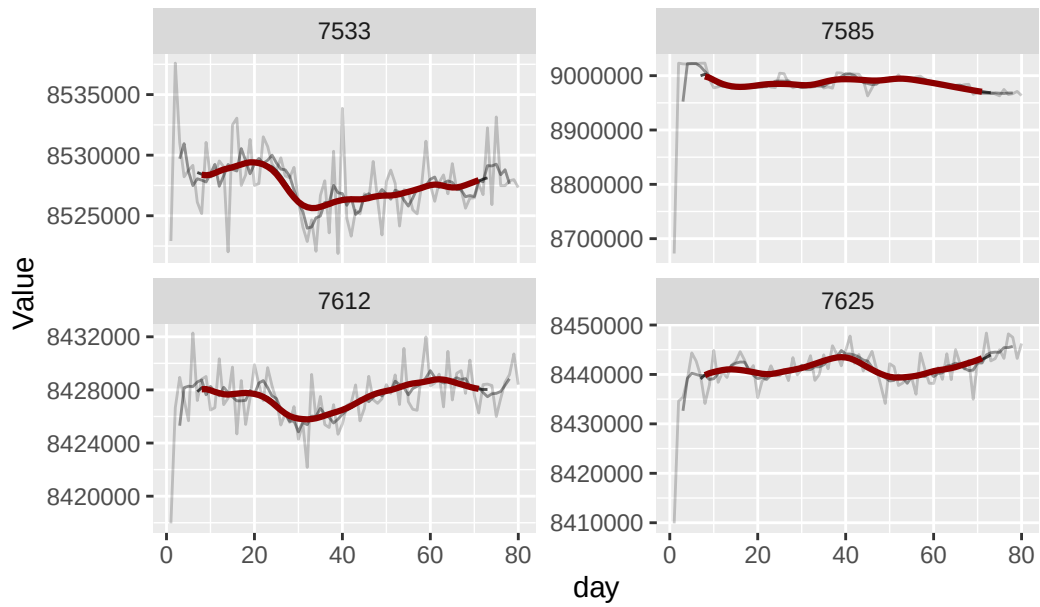
## Progressive MA Smoothing (5 ... 10 ... 4) — Page 2



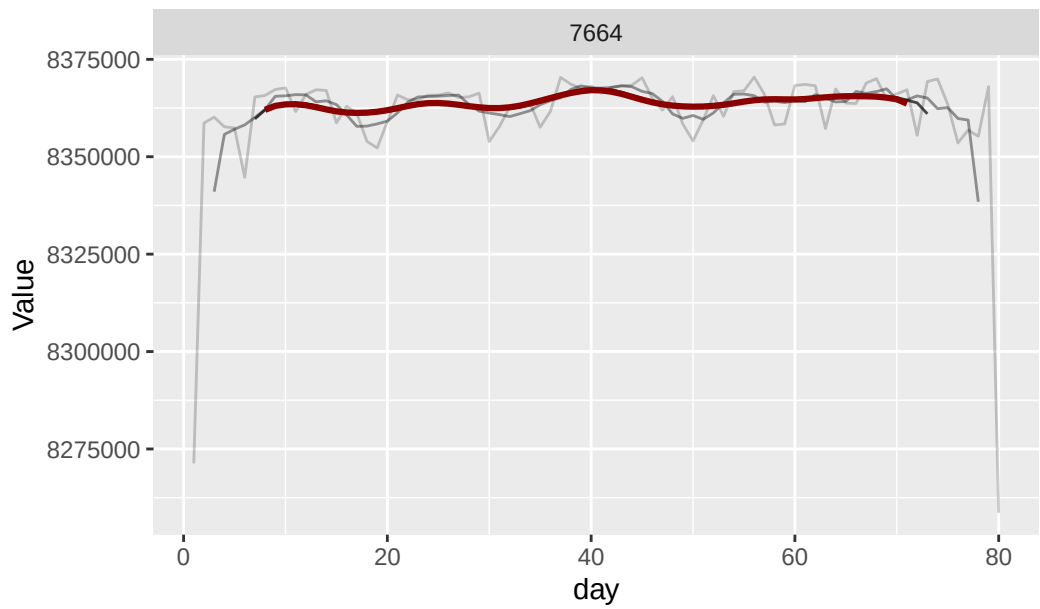
## Progressive MA Smoothing (5 ... 10 ... 4) — Page 3



Progressive MA Smoothing (5 ... 10 ... 4) — Page 4



Progressive MA Smoothing (5 ... 10 ... 4) — Page 5



### 4.5.3 Summary of Third Smoothing Stage

The number and proportion of remaining series that become stationary after the third smoothing stage are calculated.

```
# A tibble: 1 x 3
  total stationary_after_ma3 proportion
  <int>          <int>          <dbl>
1     17             0             0
```

## 5. Interpretation: Moving Toward a Calibration Threshold

The progressive smoothing analysis is intended to determine whether the apparently non-stationary white-noise series contain an underlying, slowly changing calibration signal. If additional smoothing does not produce stationarity, the remaining variation may represent genuine long-term calibration behavior rather than short-term random noise.

The next step is therefore to examine the residual error around the final smoothing approach rather than continuing to smooth indefinitely.

### 5.1 Residual Analysis Around the MA(10) Signal

For the remaining series, the residual is defined as:  $\text{residual} = \text{observed value} - \text{smoothed value}$ . The residual standard deviation and variance quantify the amount of short-term variation around the underlying smoothed signal. The range and 80th percentile of the smoothed signal are also calculated as potential measures of the operating calibration range.



## 6. Next Analytical Question

The analysis now provides the components needed to investigate a future recalibration rule:

the underlying calibration behavior estimated through smoothing, short-term residual variability around that behavior, the amount of normal calibration movement, the distribution of calibration ranges, the stationary/nonstationary classification, and detected change points for unstable series.

A proposed next step is to determine when the observed calibration behavior has moved far enough from its expected operating behavior to justify recalibration.

For the nonstationary white-noise series, the current analysis suggests using the smoothed signal as an estimate of underlying behavior and the residual distribution as an estimate of normal short-term calibration variation.

A conservative percentile such as the 75th or 80th percentile could then be evaluated as a candidate threshold. The appropriate percentile and the exact recalibration trigger still require validation against the actual calibration performance and desired false-alarm/recalibration tradeoff.

For non-white-noise, nonstationary series, the next step is to investigate whether smoothing can reveal piecewise-stationary periods. Those periods can then be evaluated to determine when the calibration behavior changes enough to warrant recalibration.